

Experiment – 03 (HomeTask)

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Subject: Advanced Database Management System

Question 1: Ad Campaign ROAS

Source: [DataLemur — Ad Campaign ROAS — https://datalemur.com/questions/ad-campaign-roas](https://datalemur.com/questions/ad-campaign-roas)

Google's marketing team is analyzing the performance of its advertising campaigns. Each row of the ad_campaigns table records the amount an advertiser spent and the revenue that spend generated. Write a query to calculate the return on ad spend (ROAS) for each advertiser, defined as total revenue divided by total spend, rounded to 2 decimal places.

ad_campaigns Table:

Column Name	Type
campaign_id	integer
advertiser_id	integer
spend	decimal
revenue	decimal

ad_campaigns Example Input:

campaign_id	advertiser_id	spend	revenue
1	1	100	500
2	1	300	600
3	2	200	300
4	3	500	500

Example Output:

advertiser_id	roas
1	2.75
2	1.50
3	1.00

Solution:

```
SELECT
  ADVERTISER_ID,
  ROUND(SUM(REVENUE) / SUM(SPEND), 2) AS ROAS
FROM AD_CAMPAIGNS
GROUP BY ADVERTISER_ID
ORDER BY ADVERTISER_ID;
```

Question 2: Final Account Balance

Source: *DataLemur — Final Account Balance* — <https://datalemur.com/questions/final-account-balance>

PayPal maintains a table of bank deposits and withdrawals made by its customers. Write a query to retrieve the final account balance for each account, assuming no transactions are missing. Using account ID 101 as an example: \$30.00 was deposited and \$5.00 was withdrawn, so the final balance is \$30.00 - \$5.00 = \$25.00.

transactions Table:

Column Name	Type
account_id	integer
transaction_type	varchar
amount	decimal

transactions Example Input:

account_id	transaction_type	amount
101	Deposit	30.00
101	Withdrawal	5.00
102	Deposit	60.00
102	Withdrawal	20.00

Example Output:

account_id	final_balance
101	25.00
102	40.00

Solution:

```
SELECT
  ACCOUNT_ID,
  SUM(
    CASE
      WHEN TRANSACTION_TYPE = 'Deposit' THEN AMOUNT
      ELSE -AMOUNT
    END
  ) AS FINAL_BALANCE
FROM TRANSACTIONS
GROUP BY ACCOUNT_ID;
```